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22.Mini Assignment — Full EDA on the Penguins Dataset

Perform a complete EDA on the `penguins` dataset (already loaded) and answer the following in new cells below. This is meant to be done **independently** — use everything you've learned in this notebook!

1. Number of rows and columns
2. Number of missing values (per column)
3. Number of duplicate rows
4. A correlation matrix / heatmap of the numeric columns
5. A boxplot showing `bill_length_mm` grouped by `species`
6. A pairplot colored by `species`
7. A fully cleaned version of the dataset (missing values handled, duplicates removed)

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Load the Penguins Dataset -

```
# Load the Penguins dataset
penguins = sns.load_dataset("penguins")

penguins.head()
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex
0	Adelie	Torgersen	39.1	18.7	181.0	3750.0	Male
1	Adelie	Torgersen	39.5	17.4	186.0	3800.0	Female
2	Adelie	Torgersen	40.3	18.0	195.0	3250.0	Female
3	Adelie	Torgersen	NaN	NaN	NaN	NaN	NaN
4	Adelie	Torgersen	36.7	19.3	193.0	3450.0	Female

1. Number of Rows and Columns

```
print("Shape of Dataset:", penguins.shape)
print("Number of Rows:", penguins.shape[0])
print("Number of Columns:", penguins.shape[1])
```

```
Shape of Dataset: (344, 7)
Number of Rows: 344
Number of Columns: 7
```

2. Number of missing values (per column)

```
print("Missing Values Per Column:")
print(penguins.isnull().sum())
```

```
Missing Values Per Column:
species              0
island               0
bill_length_mm       2
bill_depth_mm        2
flipper_length_mm    2
```

```
body_mass_g          2
sex                 11
dtype: int64
```

3. Number of duplicate rows

```
duplicates = penguins.duplicated().sum()
print("Number of Duplicate Rows:", duplicates)
```

```
Number of Duplicate Rows: 0
```

4. A correlation matrix / heatmap of the numeric columns

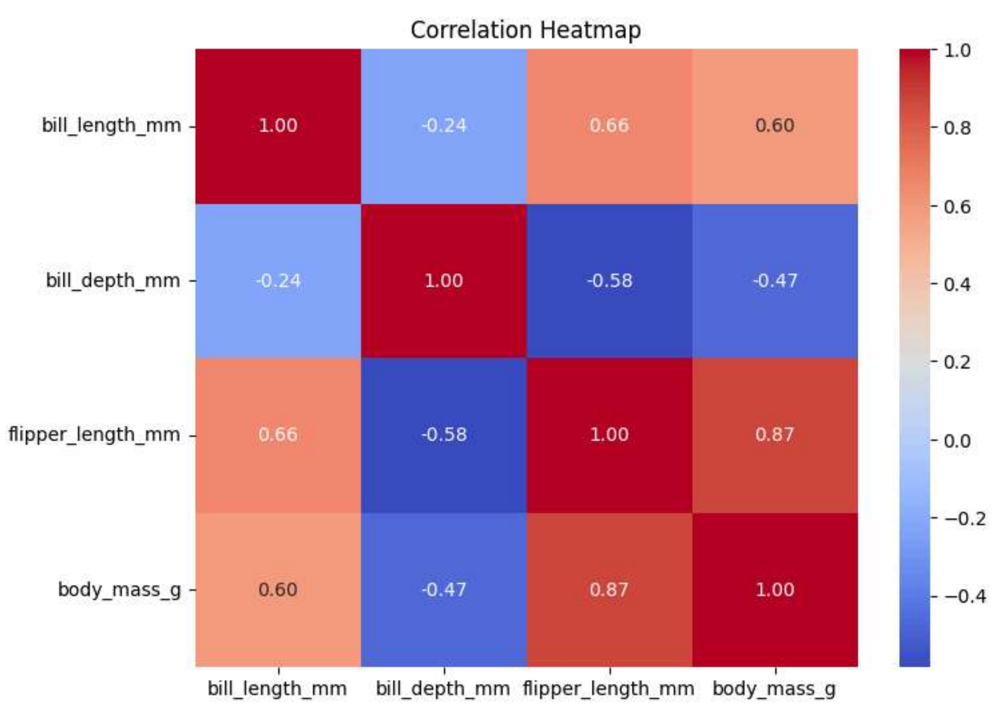
Correlation Matrix -

```
# Select only numeric columns
numeric_columns = penguins.select_dtypes(include="number")

# Correlation matrix
corr_matrix = numeric_columns.corr()

print(corr_matrix)
```

```
                   bill_length_mm  bill_depth_mm  flipper_length_mm  \
bill_length_mm           1.000000      -0.235053           0.656181
bill_depth_mm           -0.235053       1.000000          -0.583851
flipper_length_mm        0.656181      -0.583851           1.000000
body_mass_g              0.595110      -0.471916           0.871202

                   body_mass_g
bill_length_mm        0.595110
bill_depth_mm        -0.471916
flipper_length_mm     0.871202
body_mass_g           1.000000
```

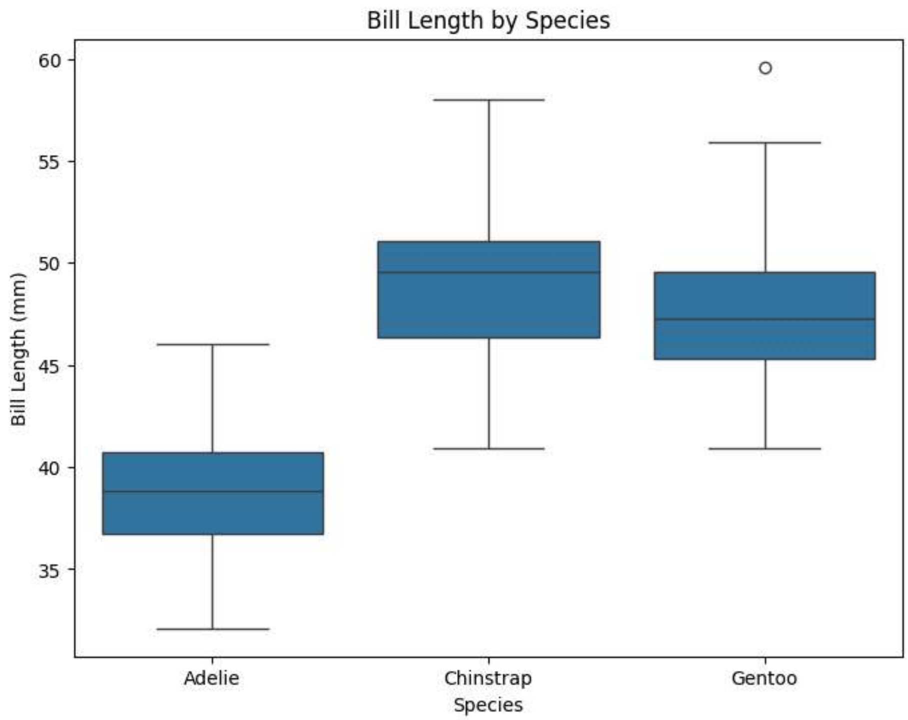
Correlation Heatmap -

```
plt.figure(figsize=(8,6))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```



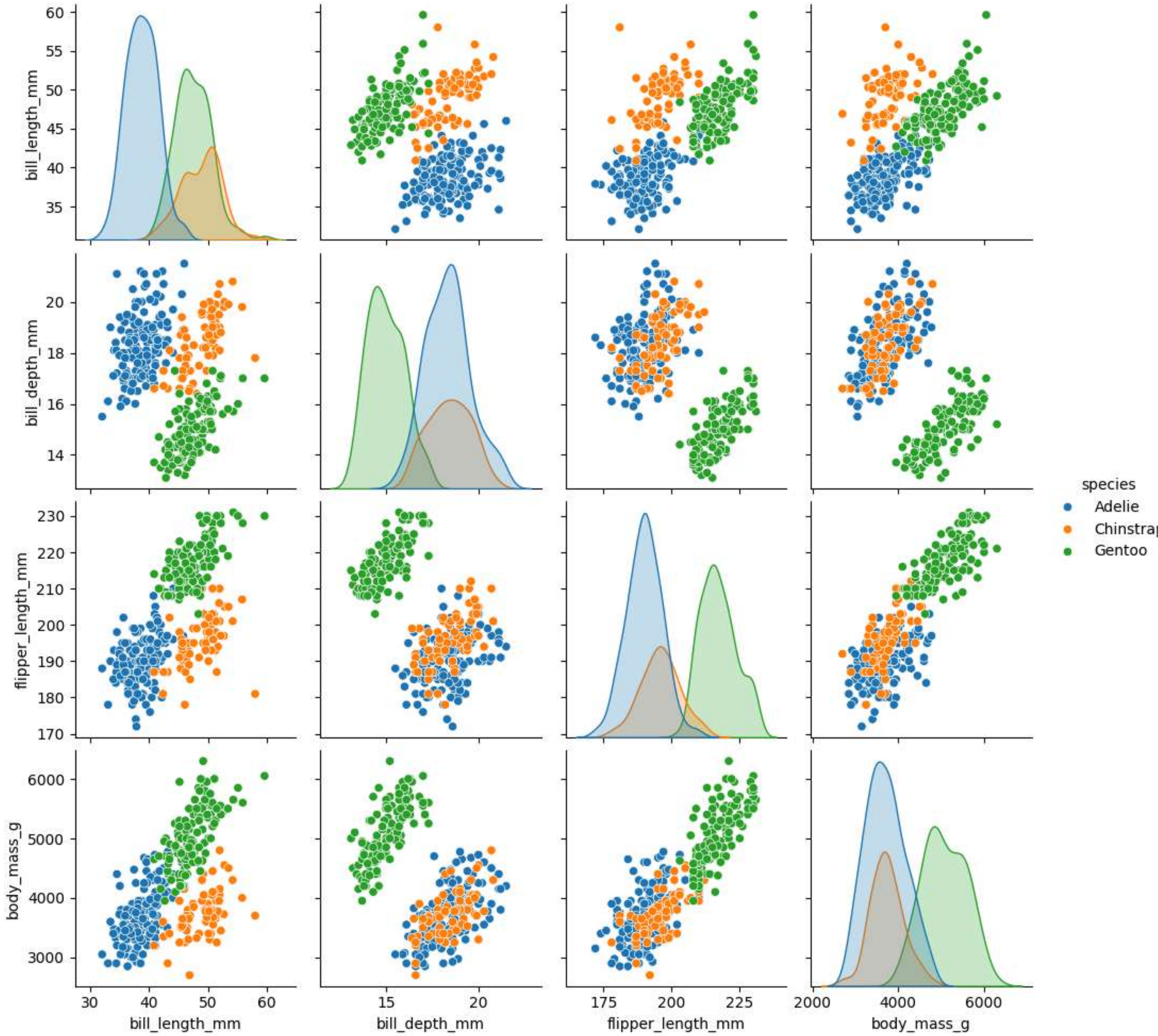
5. A boxplot showing `bill_length_mm` grouped by `species`

```
plt.figure(figsize=(8,6))
sns.boxplot(data=penguins, x="species", y="bill_length_mm")

plt.title("Bill Length by Species")
plt.xlabel("Species")
plt.ylabel("Bill Length (mm)")

plt.show()
```



6. A pairplot colored by `species`

```
sns.pairplot(penguins, hue="species")
plt.show()
```



7. A fully cleaned version of the dataset (missing values handled, duplicates removed)

```
# Create a copy
penguins_clean = penguins.copy()

# Remove duplicate rows
penguins_clean = penguins_clean.drop_duplicates()

# Remove rows with missing values
penguins_clean = penguins_clean.dropna()

print("Original Shape:", penguins.shape)
print("Cleaned Shape:", penguins_clean.shape)
```

```
Original Shape: (344, 7)
Cleaned Shape: (333, 7)
```

Verifcation -

```
print("Shape of Cleaned Dataset:", penguins_clean.shape)
```

```
print("\nMissing Values Per Column:")
print(penguins_clean.isnull().sum())

print("\nNumber of Duplicate Rows:")
print(penguins_clean.duplicated().sum())
```

```
Shape of Cleaned Dataset: (333, 7)

Missing Values Per Column:
species              0
island               0
bill_length_mm       0
bill_depth_mm        0
```